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REINFORCEMENT LEARNING

Full RL Introduction

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So far we have been looking at bandit algorithms right or the immediate reinforcement learning and so what we are going to start from today is looking at the full reinforcement learning problem right.

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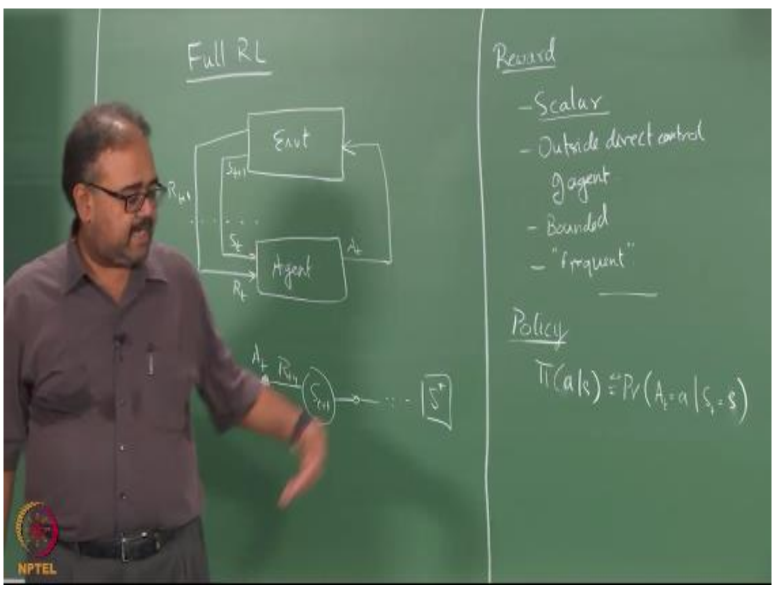

So we know little bit already about the full RL problem right so what are the complications that we are going to insert here States we already looked at states in contextual bandits right temporal dependence right the sequential nature of the problem is something that we are going to introduce here so what we are going to look at is that we will assume that there is a there is an environment okay and an agent interacts with the environment, the agent interacts with the environment at any point the agent senses the state the environment is in okay.

Then in response to that the agent takes a action which is then applied to the environment okay so the agent senses the state the environment is in that is S_t and it takes an action that gets applied to the environment and as a result of applying this action environment changes to a state S_{t+1} which then gets fed back into the agent as the current state S_t and then you keep doing this in a loop right so this could go on indefinitely or there could be a well designated stopping state right if you reach that state you stop the trials okay or a group of states I mean you can always this is this trick right.

So when I say there is a starting state and a stopping state, it could very well be a group of starting states and a group of stopping states I can always define a dummy state to which all the states go to right so I can always say that there is a single starting state and a single stopping state right so that for ease of exposition so I will keep saying that makes sense and even if there are multiple states where you will stop I can insert one dummy state and say that automatically as soon as you reach one of these states you will make a transition to the dummy state and you will stop there okay.

So there will always be a one-stop state and one start state right so that might happen and apart from this you also get a reward signal R_{t+1} from the environment okay this is like I was mentioning earlier this is a convenience a modeling convenience that we put the reward in the environment okay that is because we want to put the reward in a place which is outside the agent's direct control right agent should not just arbitrarily say okay I okay I am getting a reward of 5 now right.

So it should get this reward as a result of the behavior in the world okay it cannot just arbitrarily set its own rewards okay so that is the reason we put the reward in the environment okay so that it is outside in reality right this will be a very much more complex picture right because if you think about it I mean I have been repeatedly saying that all biological agents are reinforcement learning agents right humans are reinforcement learning agents so do you have a separate reward conduit right.

We have a reward sensing mechanism somewhere right so you only have what you sense from the world right so you look around you see whatever is whatever inputs that are coming in whether you are in you could be ingesting food, you could be looking at the outside world right somebody might be touching you on the skin whatever right so you are taking in all these complex sensory inputs and other kinds of actual physical ingestible inputs okay and then you are converting them into rewards right so that is how the rewards happen right.

So if you are hungry and you eat okay that is that is rewarding right but then you find out that you have I do not know eaten something that is gone stale right that is not rewarding it is in fact opposite of it right and if you are not hungry and somebody just gives you something to eat you just eat for the sake of you know hey give me company and then you go and eat some really bad food and things like that so there is no immediate reward apart from the company part of it but the act of eating itself does not give you any immediate reward but probably will have a consequence in the long run Yeah right so go eat some fried food just because your friend asked you to eat will have consequences in the in the long run right .

So rewards are iffy things right I mean so we are going to make it very concrete right I am going to say there is this some mechanism in the environment that gives you a reward but if you really think about it these rewards are are everywhere right in fact the last few years so Andy Barto was asking going around asking this question where do rewards come from right so when was he here 2009 yeah so I hosted him here in 2009 and one of the talks I mean at that time the talk he was going around giving everywhere was where do rewards come from okay so it was it is really an unforgettable thing yeah.

So so this is not very clear so take it with a pinch of salt when I say that the environment is going to give you a reward signal, the rewards could come from anywhere right in fact it is not even clear what is the agent what is the environment hmm so we have a lot of Robotics guys here right suppose I am having a robot learning problem right so what would be the agent what would be the environment the robot is the agent really okay let us say I give you a navigation problem or a reaching problem let us say I am trying to train a robot to reach out and pick an object on the table.

Environment is the workspace what about the arm, what about the angle of the arm is it in the environment or is it part of the agent so where is the agent then so the arm is not part of the agent the one controlling the arm is the Agent so it becomes tricky right we have to actually think about it is not like okay as a robot the robot is a agent everything outside the robot is the environment right so it is not that boundaries are not drawn as neatly around the physical entities right suppose suppose let us consider forget about robots let us say you are driving a car okay so who is the agent what is the agent and what is the environment there.

Right I mean always the right answer is depends okay but but what does it depend on right in this case the agent could be the car right the environment could be the road right so as a human being sitting and driving a car you can look at okay what is the velocity at which the car is moving okay and what is the direction which the car is moving are there pedestrians jumping in front of me right and all those things right so this could be your this could be your environment right and the agent itself is the car you know I am controlling the car on the other hand the environment could include the car right and the agent could be the human sitting there and you can think of okay.

How much how much am I pushing my foot down on the pedal which pedal should I be keeping my feet on first case right and then should I how much what torque should I apply to the steering wheel right where should I be turning my head to look at right should I be looking at the mirror should I look at front right or should I be looking down at my phone right so all of these things decisions that you have to make right while you are driving right so there is this interface now the agent environment interface becomes the human and the car right or it could be even more right you can go further back you can even go further back into the human right thinking about okay which okay I am talking about the motor cortex right.

So which neurons in my motor cortex should fire now so that my arm twitches enough so that I turn the wheel this much right so which neurons should fire in my motor I mean the autonomous nervous system so that my leg actually pushes down with some kind of pressure right so I could I could think of depending on the kind of problem that I am solving right I could think of drawing this agent environment boundary anywhere right I mean not arbitrarily anywhere but you can think of multiple places where the same set of interacting entities are there right but you have very different kinds of modeling that you could do.

Okay it is not very clear when I say there is an agent there is an environment do not expect people to come and give you okay this is the agent here is the environment now come up with a reinforcement learning algorithm that will solve this problem right so typically you will have to think about this figure out what is a right abstraction that you would need right before you go start a solution for this right so this is something which you have to keep in mind nobody is giving this to you apriori right so having said all of this one other way of thinking about how this interaction proceeds right.

So modulo all the caveats I have been giving to you there is an agent okay it senses the state of the environment it takes an action okay and then that causes a change in the state of the environment right so having said all of this you can think of the interactions as proceeding like this so I have S_t right I sense that I take action A_t okay that causes new state S_{t+1} and a reward R_{t+1} so this is a slight quark of the book itself right where the reward corresponding to the state and action at time t is labeled as R_{t+1} okay so this is a quirk in the book and the reason they say is typically the next state and the next reward are determined at the same time.

And therefore it makes more sense to label both of them with t+1 okay so one thing here in the bandit case I was making a distinction between n and t okay I am going back to using t here but the t could be discrete time as well in fact for most of the course t is going to be discrete okay in fact t need not even be time, t need not even be time okay so t is steps so what are you talking about here typically these are my decision epochs right when I say t = 1 that means I am taking my first decision when you say t = 2 that means I am taking my second decision right this is typically okay.

Sometimes I am also interested in cases where I do not take any decisions but the state keeps changing over time but for most parts when I say when I advance t by 1, that is typically at the point where I am required to take the next decision okay so later we will actually look at a still a discrete-time case but where decisions need not happen at every time tick okay so we will look at that case as well later but not immediately okay for most part of it assume that t is a decision tick ok not a time tick when t goes from t to t+1, that means have taken another decision the decisions could need not necessarily take the same time.

For example consider the game of chess right so each move that you make I mean so initially you will be making moves very rapidly right because you are playing from a standard opening play or something and later on you actually have to study the board position and then you have to think about it and make moves right but I still count each everything as a single move right so you do not say the first few moves were all count as one-fourth of a move each or anything because you did it very fast so just exactly like that so each move itself could take different amounts of time right.

I am not worried about it and the intervals at which you are asked to make decisions might also be non-uniform I am not worried about it okay this is the basic setting I am having later when we talk about hierarchical RL we will start worrying about the duration of decisions if I take one action now how long will it take for that action to complete and those kinds of things we will worry about later right now we are assuming that every action takes the same time to complete so time within quotes here okay.

So that is basically it this just keeps going right so we will denote by S^+ that distinguished terminal state right so this just keeps going until I reach S^+ or it just keeps going indefinitely if there is no distinguished terminal state okay any questions so far, the setup is clear right so what we are going to do okay next thing I want to talk about is the reward itself so I am going to be very religiously following chapter 3 in the book right and I did ask people to read ahead to Chapter three how many of you actually read chapter three, four my god Subho you did you put your hand up.

Okay then no that is not counted so 4 people yeah sorry you are busy doing the assignments is it hey release the assignments from chapter 3 as well so from now onwards what we will do is before I start the chapter we will release the assignments will that help written assignment of course not the programming assignment yeah so when is when is the second written assignment due Friday is it so if you release some questions today they will be able to finish it before Friday night I only promised I will not release it on Thursday.

Right I promised I will not release questions on Thursday right I did not say anything about Wednesday.. Varun is not giving any opinion you were just thinking about okay they will not do anything to him I am the one who will get beaten up okay so we will extend the deadline to Monday and then give them additional assignments today yes so the people who read ahead so what does the chapter say about reward functions does it say anything about rewards yeah anything else about the reward so classically reinforcement learning assumes that the reward is a scalar quantity alright so the rewards are so the rewards are scalar and it is outside the direct control of agent.

That is why I told you we put it in the environment yeah we will come to that right so what does that mean that the agent cannot arbitrarily set the reward it has to do it through interaction with the environment okay so that is essentially the two main things that we need so the more important thing here is the assumption that the rewards are scalar right so what does it mean it means is just a number right so it might look that it is a pretty weak model for the rewards but if you remember we talked about a wide variety of applications of reinforcement learning in the first lecture and in all of those applications that we went through we were actually looking at scalar robots.

Right so it does turn out that there is a wide variety of settings where you can use the scalar rewards and get away with it right but there are cases where you might have to make some kind of calls about trade-offs between how much importance you want to give to one event versus another so for example let us again take the reaching tasks that we had right so the so when I want to reach out to an.. how would you set the rewards here once I grasp it I will give a reward right I can give you some reward like I can give you a reward of 10 or something when I grasp the yeah but then I need to optim..I need to get there right.

So I what I am trying to do is optimize my reward function if you remember the whole problem in the reinforcement learning is to well this is something I mentioned in the very beginning the problem in RL is to behave in such a way so that you get as much as reward as possible in the long run right so if I do not give you any reward function right I will just be flailing around right I could be happily moving waving my hand around not doing anything with regard to grasping the object right.

So I need to give some kind of a reward either for grasping the object or how else would you do this negative reward for every time step I keep flailing around right it is like you are in constant pain right and the pain will stop once you hold this right because the episode ends and you stop getting -1s there right so it is like somebody keeps poking you with a pin all the time hmm keep go go until you reach this is how we train yeah any any any animal that draws a vehicle right you keep pushing it in the side until it actually reaches the destination right.

I mean humans are very cruel man I mean anyway so that is how we do it right so but then is that sufficient so you do not really want to strain your hand right so I mean there are joint limits right so you really do not want to come up with a grasp grasp that looks like this right this is also valid I have caught the phone right so you do not want to do that right you want to give some kind of a.. you know additional pain right if I do things like this right so if in fact for me it actually was painful right but so you want to do this communicate this to the bot right how much additional pain do you give it.

So grabbing this thing let us say gives you a reward of 10 right and doing that gives you a reward of say -100 so what is going to happen you are going to be very cautious right in trying to grab it right but if grabbing this gives you a reward of 100 right and doing that gives you a reward of -0.1 right you would still be cautious because I am being optimal right so 99.1 is worse than 99.5 so I will still be cautious about avoiding those but I will not go to to significant extent to do that because I am still getting my -1's right so the longer I take to get to the get to a grasp right the more -1 I am going to get right and the better form the grasp is, the fewer -0.1s I am going to get.

So there is some trade-off I will form at some point right but where the trade-off point lies depends on the magnitude of the rewards I am going to give you as well so if I am going to give you a -1000 for doing something like this right then I will be more careful right so these are all trade-offs right when you design the problem but so how do you set these reward functions well understood problems like grasping or reaching or robotics in general right you can look at the problem that you are trying to solve and try to figure out okay which are the undesirable configurations okay,

So very expensive robot so I do not want to break it so let me make sure that all these other things have a highly high negative values right and versus a cheap robot or what I really want to do is get there as soon as I can okay do not mind throwing 300 robots at it because each robot costs only 1 rupee right then I can just not worry about crashing and burning a few along the way right so it depends on the domain right so there is still a significant trade off that you will have to think about right so even though I am saying it is a scalar quantity there are instances where I will have to kind of put multiple outcomes right multiple outcomes on the same scale right which makes it a little tricky just like in this case I mean there is an outcome associated with success which is grasping it and there is outcome associated with well not really failures but outcomes associated with undesirable events along the way.

So I have to put all of these on the same scale and I have to do it might have been more convenient if you could have thought of okay 100 for that ok, -5 for this but these are just things that you need to avoid you know if there is some other way I can communicate these things to robot then that might be a better mechanism it turns out it is hard to come up with an appropriate optimization set up for doing this so the scalar reward thing A) is pretty powerful and B) kind of gels in some sense with how we also do decision-making at the end of the day we are very good at judging trade-offs right.

So whenever as soon as you start thinking of doing these kinds of trade-offs in some sense you are putting these on a same scale right the putting them on the same scale because they only then you can think about trade-offs so in that way also it seems to be natural and the third thing is having a single scalar value makes it easier for us to optimize the solution right we can always get a good optimal solution this is for various reasons the scalar values are fine right so I think that is one of the homework questions which ask you to think about scalar values a little bit more I mean think and reason about it.

And so what else do we need the reward to do well mathematically we need the rewards to be bounded when we are talking about biological systems that is not really a criterion because I mean it has to be physical right so you cannot have an unbounded neural spike in your brain for corresponding to a reward right so you may just fry your brain right so but when talking about mathematical settings you really want the reward to be bounded right and so basically that is it so one other thing later on we would like the rewards to be frequent I will explain this later like once we start looking at learning algorithms and stuff but this essentially means that it should not be that I get rewards say once in a million time steps.

There is very little very limited literature in this right so the one alternative which people have explored is to look at vector-valued rewards where each component of the reward corresponds to a different thing that you want to achieve and these are called typically multi-criteria RL right there are very few papers maybe five or six papers out there which talk about multi criterion reinforcement learning in fact there is a corresponding optimization thing also called multi-criterion optimization right.

So where people talk about different notions of optimality even so what exactly is optimal right so optimal is where the objective function you reach a minima of the objective function but now if you are going to look at it in multiple dimensions so you may end up reaching some kind of saddle points also right so what exactly is the optima in this case so people talk about those kinds of characterization.

So there is more work in multi-agent settings in looking at more richer reward functions than in single agent RL right yeah very little out there and it will be interesting to come up with some kind of qualitative feedback as well instead of looking at this kind of a quantitative feedback and if you like that kind of work there are there is some kind of work in fuzzy RL where the feedback signal is a fuzzy signal not much again very little and there is one more thing I want to mention yeah but all of those are scalar I mean there are all kinds of work on like instead of looking at the reward but look at the perception of reward and so on so forth.

But all of these are still scalar quantities and they are not vector valued quantities but it is a much richer notion of feedback than what you would is typically a reward function yeah so what we mean by frequent is that you should not be getting rewards like once in a million time steps or something then you will never learn anything right you are trying to optimize rewards and regardless of what you do for a million time steps you get nothing right that is too long a horizon for you to learn anything over right so typically you expect more reward to come to you along the way okay.

So that is what is rewards so policy so what is it so I am just talking about different components of a reinforcement learning system okay so we talked about states and actions right basically the agent and the environment now you're talking about the reward okay the next thing you want to talk about is the policy so what is the policy no 0 is non informative right I really cannot change my unless I have gotten a -1 somewhere I do not know if 0 is good or bad right I mean I really cannot change anything based on 0 but that is a good point yeah.

So when we say frequent rewards I do mean frequent non zero rewards right so we know what a policy is right all of us know what policies are so what is a policy so I am going to denote a policy as this as a conditional thing from now on so what is this is essentially the probability that right the probability that at time t the agent will take action a given that at time t the state is small s.

So ideally I should have a t there right so $\pi_t(a|s)$, is a probability that at time t I will take action a given that at time t the state was s okay

$$\pi_t(a|s) = Pr(A_t = a|S_t = s)$$

so normally for most of the material that we cover right I will be looking at what are known as stationary policies so what is a stationary policy right a stationary policy is something which is essentially $\pi_1 = \pi_2 ... = \pi$ so regardless of the value of t I will be using the same probability so the probability that let us say my state is 1 right probability that when give me some other symbol to use as state let us say my state is l okay I do not think I used l for anything so far right we have in the proofs you use l or is it yeah let us say okay fine let us use it one then people can all of you are smart enough to figure out this right.

So let us say state is 1 right so I do not care whether it is 1 that time one whether s_1 is equal to 1 or s_{25} is equal to 1 the probability that I will take action A_1 will be the same right the probability I will take action A_2 will be the same so essentially what I am saying is.

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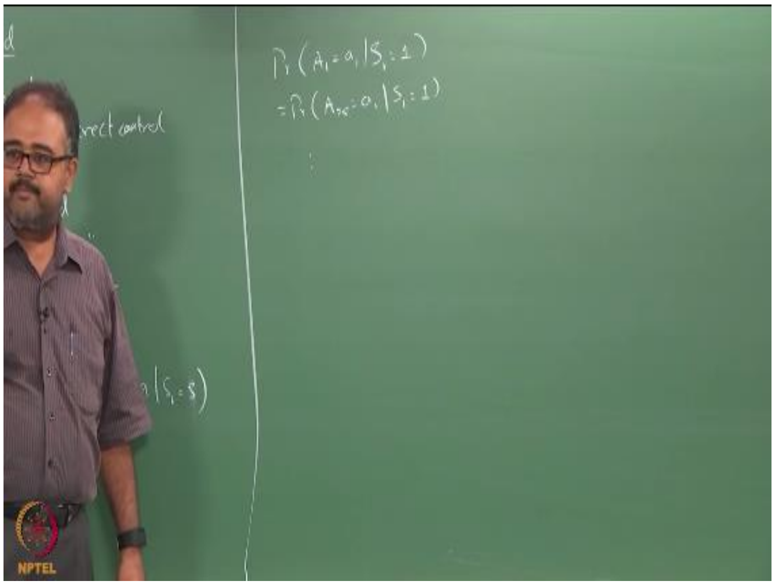

Probability that A_1 equal to a_1 given S_1 equal to 1 is equal to probability that something like this okay right so this is what I mean by stationary that is what I am saying here so if I want to explicitly talk about non-stationary policies okay I will use the π_t notation okay if I am talking about stationary policies I will just use the π notation.

Okay so what is my policy the what is the goal in trying to come up with a policy no see we have not even gone there right I am just defining things right a non-stationary policy is just a collection of functions collection of probability distributions one for each time step right a stationary policy is a single well it is actually also a set of probability distributions one for each state right a non stationary policy is a collection of such sets one for each time step as well as one for each state that is basically it how you derive how you derive a non-stationary policy or how you learn it from data and other things they are independent questions I am just talking about its simple definition of the quantity when you learn your goal should be a stationary policy right the end goal can be a stationary policy but if you have a stationary policy there is no subsequent learning we will come to that right so there are different time scales at which you can learn right.

So I can have a stationary policy run the full episode using the stationary policy and then take the rewards I get and then use that to update my parameters and come up with a new stationary policy right so these times I am talking about the t here is essentially the index over a single interaction okay I am not talking about multiple interactions over the single interaction right suppose I start from some S_0 I come and end at S^+ over that interaction my policy stays fixed that seems to be a reasonable assumption right.

So you could think of having states learning a sequence of stationary policies so what is my goal in learning a policy here is to learn something that will maximize the reward that I get right but not individually but something that will maximize it in the long run okay so what do I mean by that suppose I am riding a cycle right so suppose you leave your hands and say they look at me or something like that might give you a temporary kick you know I feel really good about myself but then you might the next minute you might ram into another vehicle or a tree or a building whatever right and get injured right.

So that is something that gives you a well depends but hopefully that is something that will deter you from doing crazy things on a cycle the next time around right so the short-term reward was just feeling good about you for one minute and the long term penalty was getting into a crash or whatever right so these kinds of things are odd or even simple things like in chess you know the first thing kids they try to capture the Queen and whole bunch of other things and then as a result end up losing the game right so these kinds of things ok so do not look for short-term rewards you should look for something that maximizes your rewards in the long term.

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